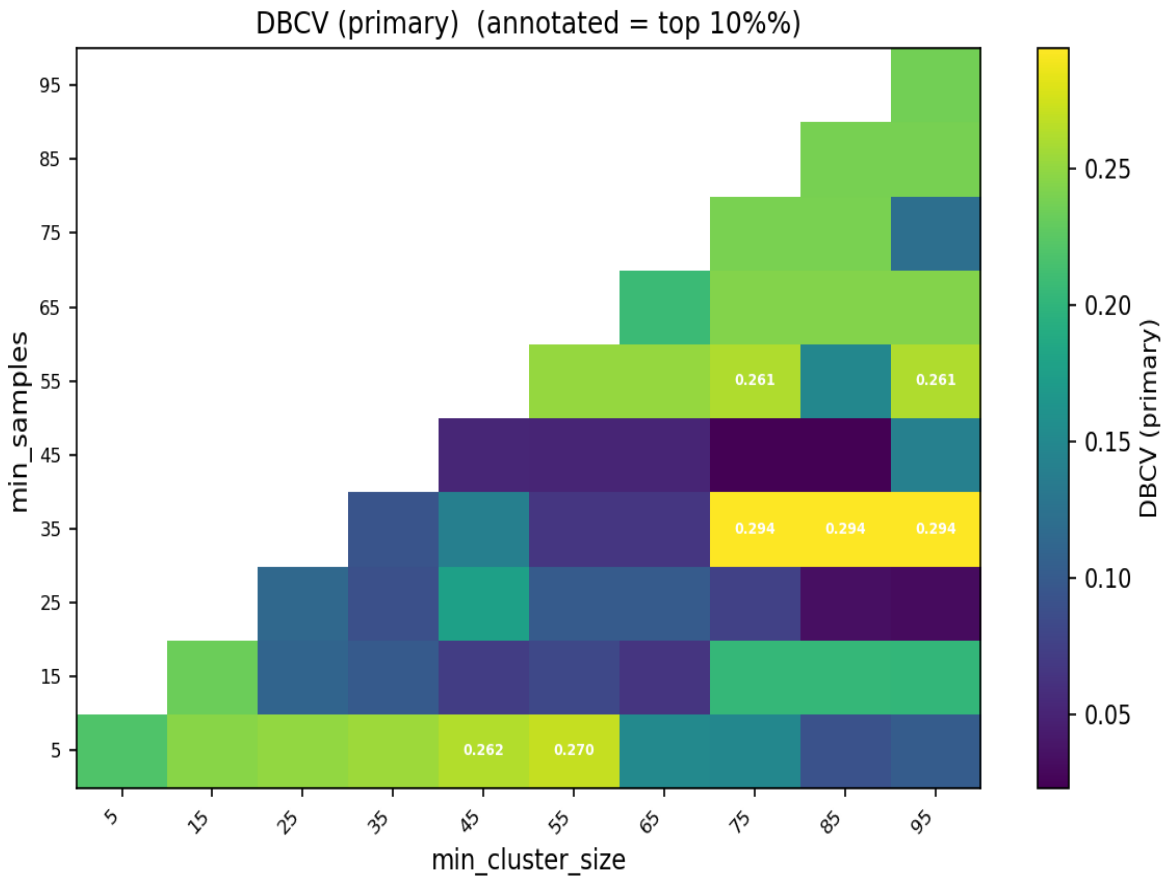
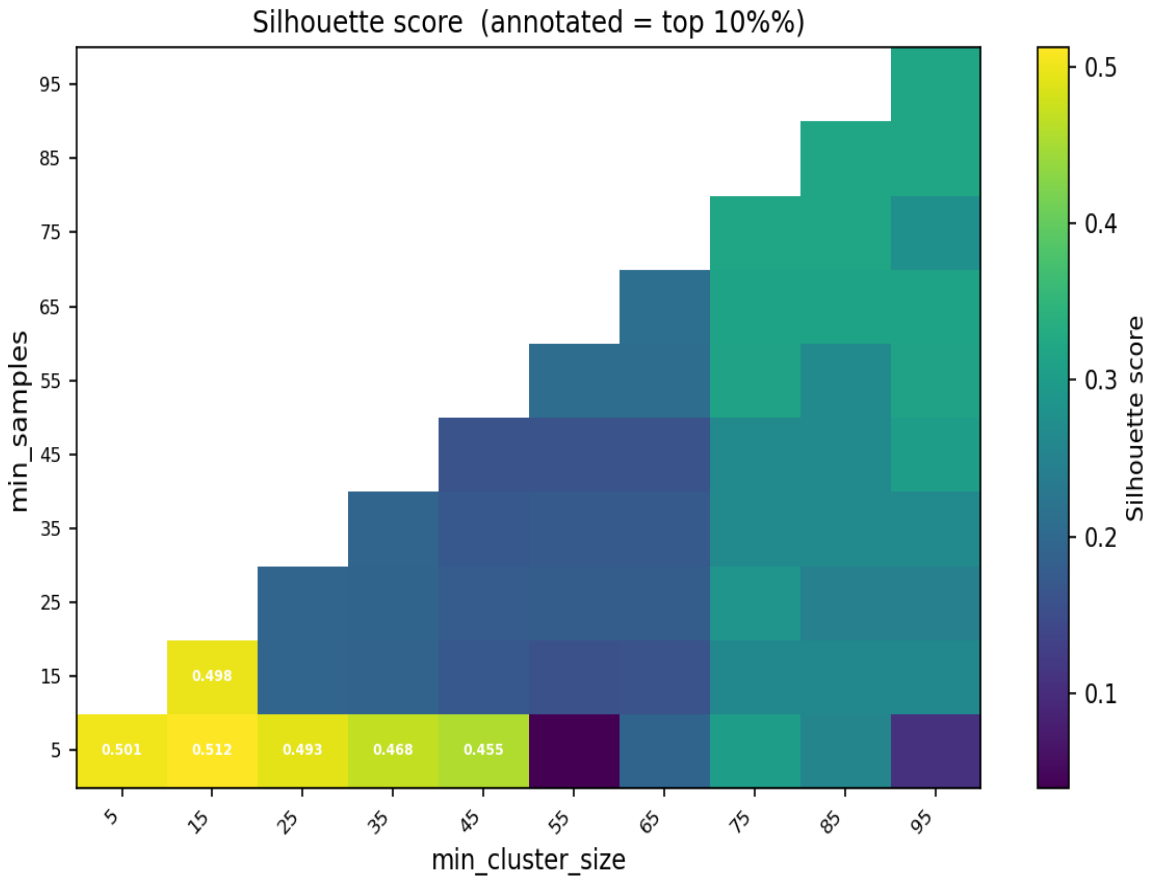
FragTree — Coarse Grid Search Report

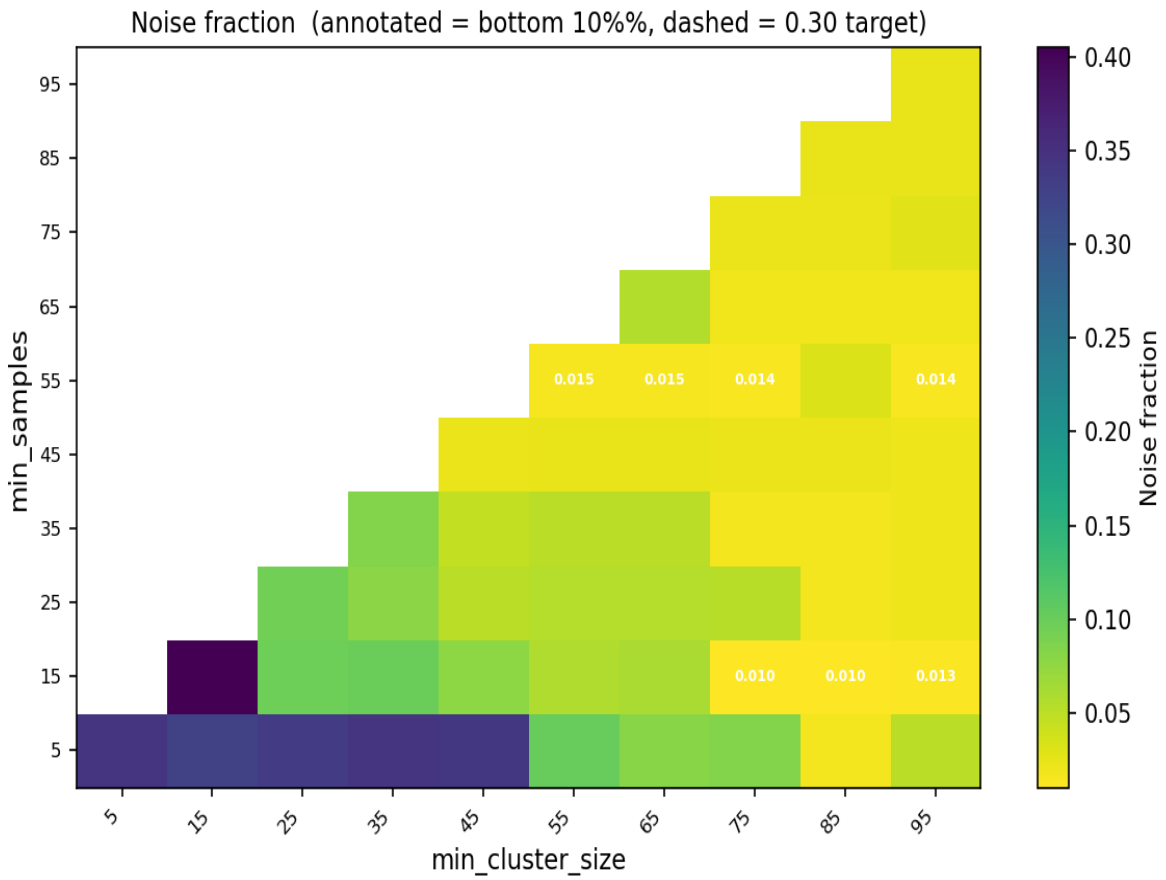
Generated: 2026-04-14 11:15:54 | n_molecules: 30,494

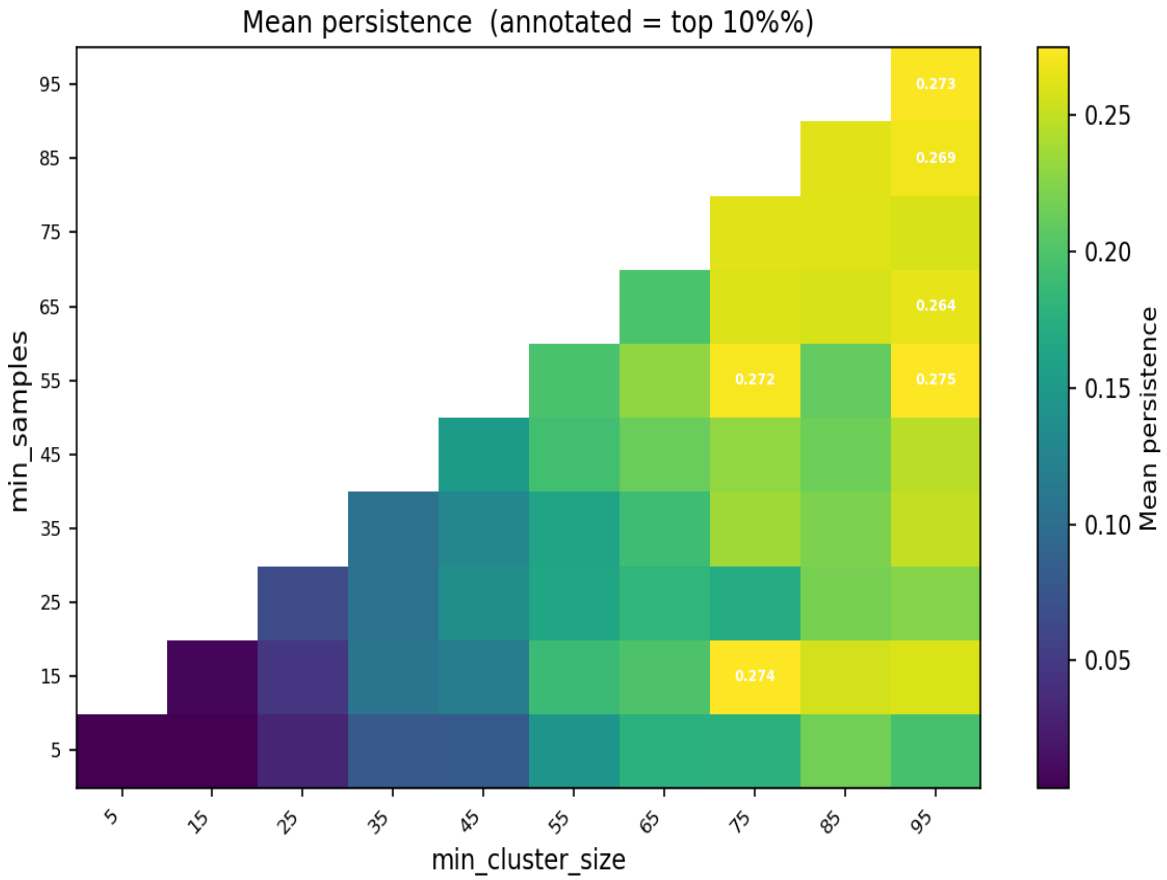
Parameter Sweep Heatmaps

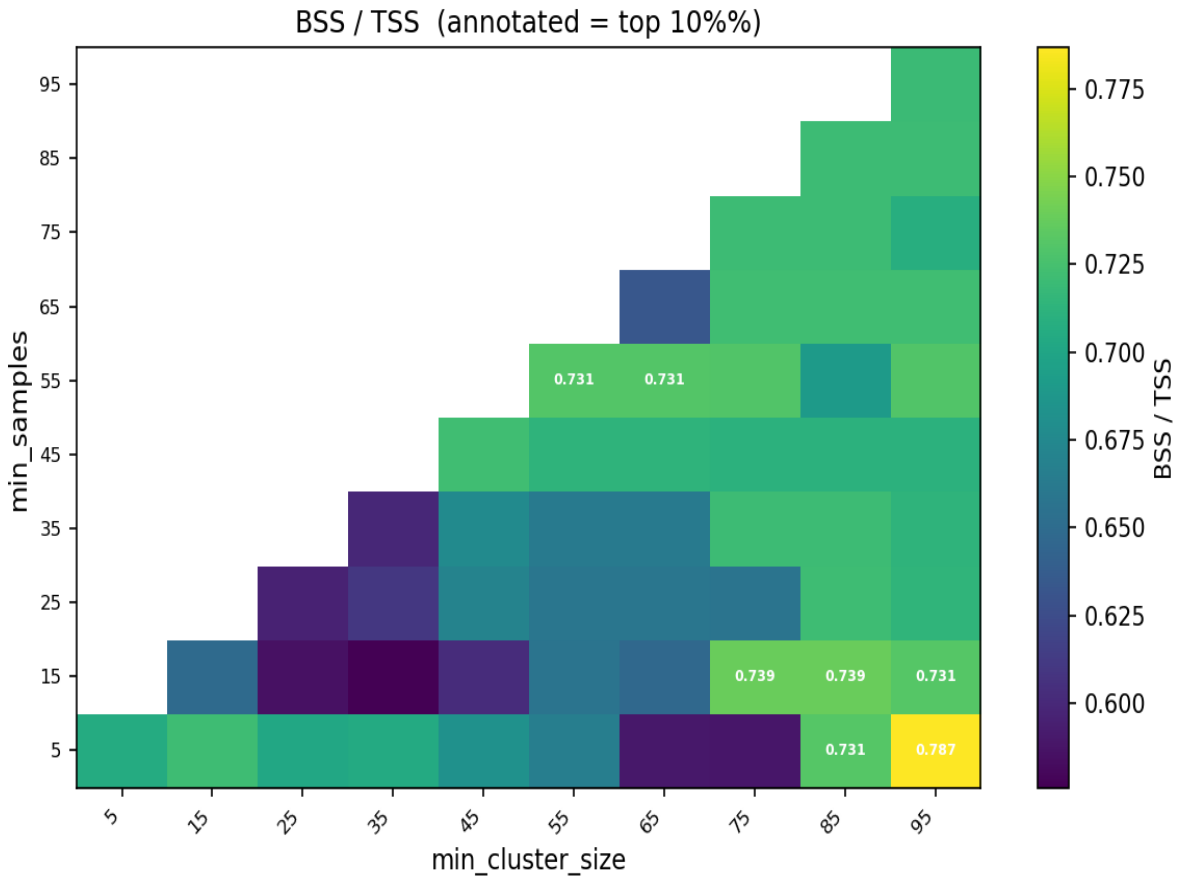
Each cell shows the metric value for a (min_cluster_size, min_samples) pair. Maximize DBCV, silhouette, mean persistence, and BSS/TSS. Minimize noise fraction (keep below 0.30). Cells with values in the top 10% are annotated with their score.




Silhouette score (annotated = top 10%%)
min_samples
min_cluster_size
Silhouette score
0.501
0.512
0.493
0.468
0.455
0.498


Noise fraction (annotated = bottom 10%%, dashed = 0.30 target)
min_samples
min_cluster_size
Noise fraction
0.015
0.015
0.014
0.014
0.010
0.010
0.013
5
15
25
35
45
55
65
75
85
95
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05


Mean persistence (annotated = top 10%%)
min_samples
min_cluster_size
Mean persistence
95
85
75
65
55
45
35
25
15
5
0.273
0.269
0.264
0.272
0.275
0.274
0.25
0.20
0.15
0.10
0.05


BSS / TSS (annotated = top 10%%)
min_samples
min_cluster_size
BSS / TSS
95
85
75
65
55
45
35
25
15
5
0.731
0.731
0.739
0.739
0.731
0.731
0.787
0.775
0.750
0.725
0.700
0.675
0.650
0.625
0.600

Top-10% Summary Table

A parameter pair is included if it falls in the top 10% of ANY metric (or bottom 10% for noise fraction, where lower is better). Sorted by DBCV descending.

cs	ms	DBCV	Silhouette	Noise frac	Mean persistence	BSS/TSS	N clusters
75	35	0.2939	0.2637	0.017	0.2357	0.7207	10
85	35	0.2939	0.2637	0.017	0.2216	0.7207	10
95	35	0.2937	0.2644	0.020	0.2496	0.7128	9
55	5	0.2703	0.0400	0.101	0.1435	0.6659	49
45	5	0.2623	0.4552	0.341	0.0806	0.6829	151
75	55	0.2614	0.3117	0.014	0.2723	0.7291	7
95	55	0.2614	0.3117	0.014	0.2748	0.7291	7
35	5	0.2540	0.4683	0.345	0.0793	0.7048	195
55	55	0.2511	0.2088	0.015	0.1964	0.7308	8
65	55	0.2511	0.2088	0.015	0.2300	0.7308	8
25	5	0.2496	0.4926	0.337	0.0318	0.7023	278
15	5	0.2463	0.5123	0.329	0.0043	0.7213	435
95	65	0.2437	0.3143	0.018	0.2644	0.7225	7
95	85	0.2389	0.3175	0.023	0.2693	0.7204	7
95	95	0.2363	0.3177	0.023	0.2730	0.7197	7
15	15	0.2326	0.4983	0.405	0.0073	0.6492	269
5	5	0.2193	0.5013	0.346	0.0035	0.7057	977
75	15	0.2035	0.2601	0.010	0.2740	0.7389	10
85	15	0.2035	0.2601	0.010	0.2559	0.7389	10
95	15	0.2020	0.2607	0.013	0.2590	0.7310	9
95	5	0.1017	0.1080	0.051	0.1956	0.7868	16
85	5	0.0911	0.2541	0.017	0.2175	0.7311	11